

Computer Science Portfolio: Design, Implementation, and Reflection

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Live portfolio: pattson22.github.io
Repository: github.com/pattson22/pattson22.github.io

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1 Introduction

This report documents the design and development of my computer science portfolio, created as part of the B201 Computer Science Lab module at GISMA University of Applied Sciences. The portfolio is a personal website, hosted on GitHub Pages, that presents my skills, projects, and credentials to a specific audience: hiring managers recruiting for working student (Werkstudent) cloud engineering positions in Germany.

I chose that audience deliberately because it shaped every decision I made. When I looked through real job listings on platforms such as Hays and StepStone, the same requirements

appeared again and again: AWS, Terraform or other Infrastructure as Code tooling, Docker, CI/CD pipelines, Linux fundamentals, and a candidate who can document and explain what they have built. A portfolio aimed at that market therefore cannot simply list technologies. It has to prove, with evidence, that I can build, automate, secure, monitor, and clearly communicate cloud infrastructure.

2 Portfolio Design and Structure

2.1 Website Structure

The website is a single-page application divided into clearly labelled sections that a visitor scrolls through in a deliberate order: a hero introduction, an “About” section, certifications, featured projects, a live GitHub feed, technical exercises, and contact details. The hero section states who I am and what I do in one glance, with a short animated terminal that signals my comfort in a command-line, infrastructure context.

2.2 GitHub Repository Structure

The repository is organised so that anyone can navigate it without explanation. The website files (`index.html`, `style.css`, `script.js`) sit at the root, as GitHub Pages expects. Supporting material is grouped into purpose-named folders: `/assets` for images and architecture diagrams, `/cv` for LaTeX CV and compiled PDF, `/exercises` for the three technical demonstrations, and `/report` for this document. A single top-level `README.md` ties everything together.

2.3 Design Decisions

I chose a single-page layout because recruiters typically spend very little time on any one portfolio. A single, scannable page removes navigation friction and lets a reader reach the projects in seconds. I added a dark and light mode toggle for accessibility and personal preference; the choice is saved to `localStorage`. I selected an AWS-orange accent colour deliberately to visually tie the site to the cloud platform I specialise in. I hosted on GitHub Pages because it is free and integrates directly with the repository.

3 Tools and Technologies

I kept the technology stack deliberately lean, choosing tools that match the job rather than the most fashionable option.

- **Frontend:** HTML5, CSS3 (custom properties, Flexbox, Grid), vanilla JavaScript.
- **Hosting and VCS:** Git, GitHub, GitHub Pages.
- **Documentation and CV:** LaTeX (Overleaf).
- **Diagrams:** draw.io (diagrams.net).

4 Portfolio Content

4.1 Featured Projects

I curated five projects and ordered them by how directly they map onto German cloud engineering requirements.

1. **Multi-tier Cloud Application** — three-tier AWS environment provisioned with Terraform, containerised with Docker, deployed via GitHub Actions.

2. **Highly Available Crypto API** — multi-AZ design with ALB and Auto Scaling.
3. **MLOps API Infrastructure** — reproducible containerised deployment for ML serving.
4. **Group Chat Application** — full-stack real-time concurrency.
5. **Simple Shell (C)** — low-level Linux and OS understanding.

For the cloud projects include architecture diagrams in `/assets` and reference them here:

Figure 1: Multi-tier Cloud Application architecture (place diagram in `/assets`).

4.2 Technical Exercises

- **Bash automation script** — idempotent server bootstrap and health checks.
- **Python boto3 auditor** — flags unencrypted or public S3 buckets.
- **Networking walkthrough** — VPC with public/private subnets, routing, gateways, and least-privilege security groups.

4.3 Curriculum Vitae

My CV is written in LaTeX and available as a downloadable PDF in `/cv`. The CV highlights AWS certification, Infrastructure as Code, CI/CD experience, and the same projects shown on the site.

5 Critical Evaluation

5.1 Strengths

The portfolio's dynamic GitHub integration keeps the projects section current. The design is clean and fully responsive. The cloud-focused branding (AWS-orange accent, terminal motif) and repository organisation demonstrate operational discipline and system thinking.

5.2 Weaknesses

Current limitations include: no contact form, project descriptions that could be deeper, only one displayed certification, no blog or technical writing, no visitor analytics, and the site is English-only (no German translation).

6 Future Improvements

Planned improvements:

- Add a working contact form and lightweight visitor analytics.
- Add further certifications (HashiCorp Terraform Associate, AWS Developer).
- Start a short technical blog and expand projects into detailed case studies.
- Add a German language toggle.

7 Conclusion

Building this portfolio taught me that presenting technical work well is a discipline in its own right. The result is a focused, honest, and well-organised portfolio that demonstrates the skills a

working student cloud engineering role in Germany demands.

References

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